

ME 302: Dynamics of Machinery

Course Code	ME 302
Course Name	Dynamics of Machinery
L-T-P-C	3-0-0-3
Prerequisites	Nil
Approval	5th BoA (Mar 2015)

Objective: To introduce various techniques of dynamic analysis of machines.

Course Contents

- **Force Analysis of Mechanisms:** Concept of free body and its equilibrium, static force analysis, friction effects, forces on gear teeth; D'Alembert's principle, dynamic force analysis, force analysis of cam-follower system, equivalent dynamical systems, dynamic analysis of reciprocating engines, practical examples from actual machines. (10hrs)
- **Flywheels and Governors:** Fluctuation of energy and speed, calculation of flywheel size; Analysis of different types of governors, effect of friction, controlling force curves, sensitivity, stability, governor effort and power. (6hrs)
- **Balancing:** Balancing of rotating masses on one plane and in different parallel planes, balancing of slider crank mechanisms, balancing of in-line, V- and locomotive engines, principles of balancing machine. (7hrs)
- **Friction Devices:** Coulomb friction, belt drive system, pivots and collars, power screws, plate and cone clutches, band and block brakes. (6hrs)
- **Gyroscope:** Gyroscopic action, equation for regular precession and gyroscopic torque. (2hrs)
- **Mechanical Vibration:** Simple harmonic motion; Conservative systems; Free vibrations of systems without damping; Equilibrium and energy methods for determining natural frequency; Rayleigh's method, free vibrations of system with viscous damping, over damped, critically and under damped systems, logarithmic decrement; Forced vibrations of systems with viscous damping, equivalent viscous damping; Impressed forces due to unbalanced masses and excitation of supports, vibration isolation, transmissibility, whirling of shaft; Introduction to multi degree offreedom system vibrations: Discrete and continuous systems. (11hrs)

Suggested Books

1. Uicker, J.J., Shigley, J.E., and Pennock, G.R., **Theory of Machines and Mechanisms**, 3rd Edition, Oxford University Press 2003
2. Vinogradov, O., **Fundamentals of Kinematics and Dynamics of Machines and Mechanisms**, CRC Press 2000
3. Massie, H.H., and Reinholtz, C.F., **Mechanisms and Dynamics of Machinery**, 4th Edition, John Wiley & Sons 1987
4. Grover, G.K., **Mechanical Vibrations**, 7th Edition, Nem Chand and Brothers 2003
5. Thomson, W.T., **Theory of Vibration with Applications**, 3rd Edition, CBS Publishers 2003